2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

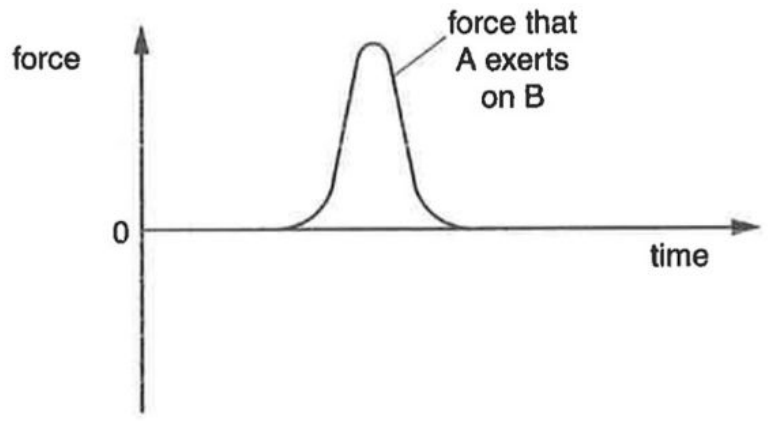
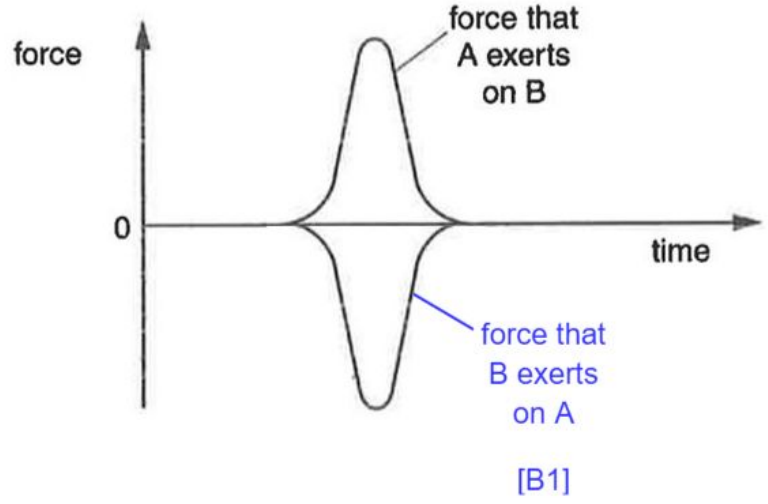


Fig. 2.1

(a) (i) Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



[B1]

Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

The forces are an action-reaction pair; therefore, they should always be equal in magnitude and opposite in direction. [B1]

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..... [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.

Calculate the change in velocity of

(i) the car,

Since the truck runs into the front of the car (as stated), this describes a head-on collision.

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

Both the truck and the car experience equal and opposite impulses (Newton's Third Law) or change in momentum.

change in velocity of the car, $\Delta v = \Delta p / m_{car}$

$= (F \Delta t) / m_{car} = 18000/1200 = 15$ m s^{-1}, [B1]

in the direction of the truck's initial motion.

change in velocity = ... m s^{-1} [2]

(ii) the truck,

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

change in velocity of the truck, $\Delta v = \Delta p / m_{truck}$

$= (F \Delta t) / m_{truck} = 18000/12000 = 1.5$ m s^{-1}, [B1]

In the direction opposite to its own initial motion.

change in velocity = ... m s^{-1} [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

Seat belts, airbags and crumple zones increase the time taken for the driver to come to rest. [B1]

For the same change in momentum, increasing the time of impact reduces the average force acting on the driver. [B1]

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..... [2]

[Total: 8]